# Joshua Aney

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## Education

**Montana State University** — Bozeman, MT
*Bachelor of Science, Computer Science, GPA: 3.96* — *May 2026*

## Research Experience

**Research Technologist** — June 2026 – Present
*CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic* — *Rochester, MN*

- Performed motor and language mapping on patients with epilepsy using computational techniques to identify anatomical locations with activity during tasks.
- Recorded data for experiments in the Epilepsy Monitoring Unit (EMU) from Stereoelectroencephalography (sEEG) leads and preprocessed the raw data from BCI2000 into a more friendly format for analysis.
- Developed a novel method for dynamic brain network identification from evoked responses.

**Undergraduate Research Assistant** — January 2026 – June 2026
*HACR Lab (Harnessing Automation in Cybersecurity Reasoning), Montana State University* — *Bozeman, MT*

- Built pipelines to produce control flow graphs from compiled binary packages for code embedding generation.
- Designed Graph Neural Network architectures using PyTorch.
- Ran machine learning workloads on Tempest, Montana State University's high performance computing system.

**Summer Research Intern** — Summers 2024, 2025
*CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic* — *Rochester, MN*

- Conducted algorithmic analysis on human sEEG data.
- Implemented signal processing pipelines including noise reduction, baseline correction, and artifact rejection.

## Publications & Presentations

### Papers in Preparation

- Aney, J., et al. *Response Lag Coupling: Identifying Response Networks Through Evoked Response Lags.* Paper in preparation for submission.
- Aney, J., et al. *A Reference and Preprocessing Guide for sEEG Data.* Paper in preparation for submission.

### Posters and Abstracts

- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects.* BRAIN Initiative Conference, 2026.
- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects.* BIONIC Symposium, 2026.
- Aney, J., Jensen, M., Kerezoudis, P., Canales-Johnson, A., Hermes, D., Baker, M., Miller, K. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects.* GATTLE Neurosurgery Symposium, Mayo Clinic, 2026.

## Teaching & Curriculum Development

**Independent Study – CSCI 491-7 Hypermedia Systems** — Fall 2025
*Montana State University* — *Bozeman, MT*

- Collaborated with a professor and a team of three students to design and develop a semester-long project template for an advanced web development course.
- Implemented core website functionality, including the MVC controller layer, authentication and authorization systems, and multi-step business workflows, for a veterinary clinic management system.
- Structured a progressive assignment sequence so students could implement new controllers by following established patterns in the codebase.

**TA – CSCI 132 (Data Structures and Algorithms)** Spring 2026
*Montana State University* *Bozeman, MT*

- Assisted students in learning core data structures algorithms (Arrays, Linked Lists, Sorting, etc.)
- Taught students how to debug in their IDEs.

## Technical Projects

**Machine Learning and AI Models from Scratch** 2024-2025

- Implemented the following machine learning algorithms from scratch without pre-built libraries: Naïve Bayes, K-Nearest Neighbors, K-Means, and DBSCAN, Multi-Layer-Perceptron, Q-Learning, SARSA, and Value Iteration.
- Implemented a First-Order-Logic engine to perform inference on knowledge bases using resolution.
- Implemented the following optimization algorithms from scratch: Backpropagation, Genetic Algorithm, Differential Evolution, Particle Swarm Optimization, and Simulated Annealing.

**Prize-Collecting TSP Solver** 2026

- Formulated the Prize-Collecting Traveling Salesman Problem as a mixed-integer linear program and solved it exactly using IBM CPLEX.
- Implemented a genetic algorithm as an alternative heuristic solver and compared solution quality and runtime against the exact CPLEX formulation.

**Bozeman Bike Trails Application** 2024

- Developed a web application to track user-ridden bike trail GPS data using .NET Razor Pages

**Hand Tracking / ASL Alphabet Detection** 2024

- Created a pipeline to collect data, train a dense neural network, and provide real time predictions for video inputs using PyTorch and Mediapipe.

## Honors & Awards

**President's List:** Spring 2022; Spring 2023, Fall 2023; Spring 2024; Spring 2025, Fall 2025; Spring 2026, Fall 2026
**Dean's List:** Fall 2022; Fall 2024

## Other Activities

**President** August 2023 – May 2026
*Montana State University Tennis Club* *Bozeman, MT*

**Section Leader** April 2025 – January 2026
*Montana State University Drumline* *Bozeman, MT*

**Tennis Coach** Summers 2023, 2025
*Rochester Tennis Connection* *Rochester, MN*

**Substitute Tennis Coach** 2022 – 2026
*Bobcat Anderson Tennis Center, Montana State University* *Bozeman, MT*